

INTRODUCTION TO DATA MINING

COURSE: DATA MINING & DATA WAREHOUSING

PROGRAM: BCA

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LECTURE AGENDA

- Introduction to Data Mining
- Importance and Evolution of Data Mining
- Real-life Applications and Industry Use-Cases
- Types of Data and Patterns in Data Mining
- The KDD Process
- Tools and Technologies Overview (Weka)
- Summary, Discussion & Q/A

WHAT IS DATA?

- Data is a collection of raw facts or figures that may not have meaning on their own.
- It can be in the form of numbers, text, images, audio, or video.
- Example: 85, 92, 76 (student marks) → These are just data.
- Once processed and organized, data becomes meaningful — this is called information.
- In data mining, data is the starting point to find useful patterns and insights.

WHAT IS DATA MINING?

- **Data Mining means finding useful information from a large amount of data.**
- 🔍 It is like **searching for patterns or hidden facts** inside a big collection of data.
- ☐ **Helps in making better decisions** by understanding data more deeply.
- 🛒 Used by companies like Amazon, YouTube, Netflix to **recommend products or videos.**
- ☐ It uses tools and techniques from **AI, statistics, and machine learning.**

Database



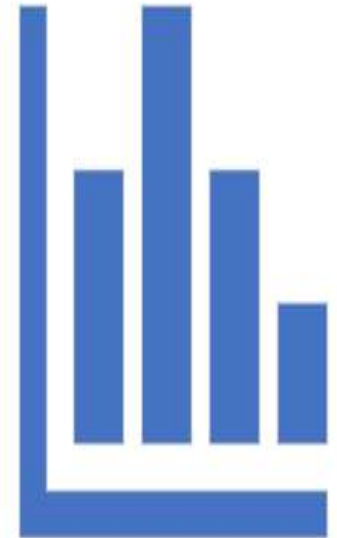
**Data Integration
and Cleaning**



File

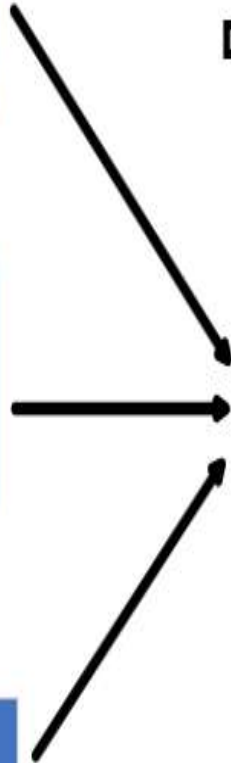


**Data Selection and
Transformation**



Analyzed Data

**Other
Sources**



STEP I – DATA SOURCES

- - Where data comes from: websites, apps, sensors, databases
- Example: Amazon gets data from your searches, orders, and reviews

STEP 2 – DATA CLEANING

- - Fix wrong or missing data
- - Remove empty entries or spelling errors
- - Clean data = better results

STEP 3 – DATA INTEGRATION

- - Join data from different places
- Example: Combine app data with website data
- - One complete dataset

STEP 4 – DATA SELECTION

- - Choose only the needed data
- Example: Use this year's sales data
- - Focus on useful data only

STEP 5 – DATA TRANSFORMATION

- - Convert data into simple form
- Example: 'Male/Female' → 1/0
- - Computers understand better

STEP 6 – DATA MINING

- - Use software to find patterns
- Example: People who buy phones also buy covers
- - Main part of the process

STEP 7 – PATTERN EVALUATION

- - Check which patterns are helpful
- - Remove false or useless patterns
- Example: Does it help with better decisions?

STEP 8 – KNOWLEDGE PRESENTATION

- - Show results using charts or graphs
- - Easy for people to understand
- Tools: Excel, dashboards

WHY DO WE NEED DATA MINING?

- Huge volumes of complex data generated daily.
- Traditional analysis is insufficient.
- Enables pattern discovery, future prediction, fraud detection, and efficiency improvements.
- Example: Social media sentiment predicting product popularity.

EXAMPLE 1: SUPERMARKET – MARKET BASKET ANALYSIS

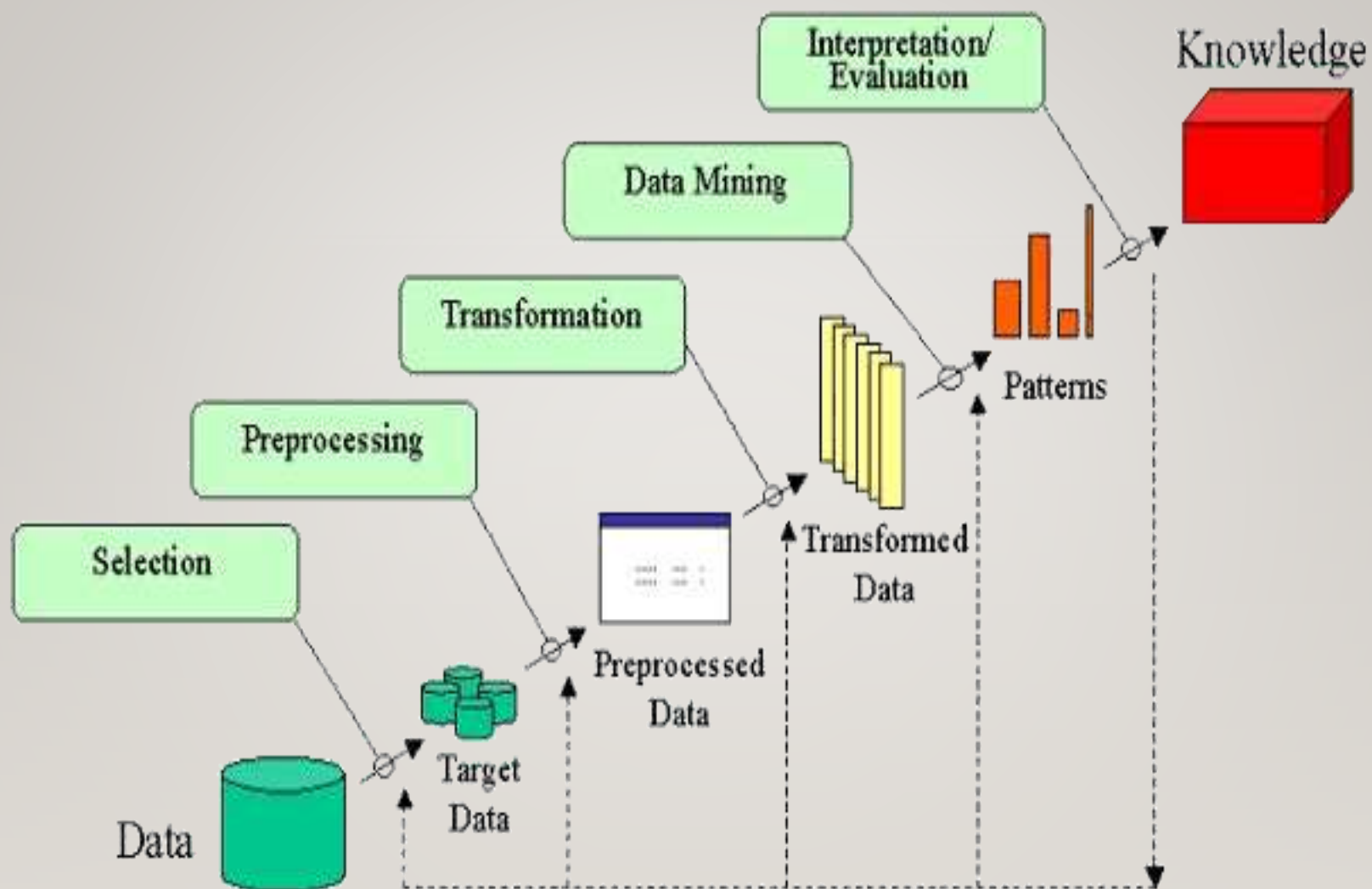
- Discovers items bought together: Bread & Butter, Shampoo & Conditioner.
- Business Use: Cross-selling, combo offers, shelf placement.
- Technique: Association Rule Mining (Apriori Algorithm).
- Metrics: Support, Confidence, Lift.

EXAMPLE 2: NETFLIX – RECOMMENDATION SYSTEM

- Analyzes viewing behavior using collaborative filtering.
- Clusters users and classifies content.
- Techniques: Clustering, Classification, Deep Learning.
- Result: Personalized experience, higher retention.

KDD PROCESS (KNOWLEDGE DISCOVERY IN DATABASES)

- 1. Selection: Relevant data chosen.
- 2. Preprocessing: Noise & missing data removal.
- 3. Transformation: Normalization, encoding.
- 4. Data Mining: Algorithms applied (Apriori, k-means).
- 5. Evaluation: Insights visualized & interpreted.



APPLICATIONS OF DATA MINING

- E-commerce: Recommendations (Amazon)
- Healthcare: Diagnostics, disease prediction
- Banking: Credit scoring, fraud detection
- Education: Student analytics, dropout prediction
- Telecom: Churn analysis, fault detection

TOOLS – INTRODUCTION TO WEKA

- Open-source Java-based data mining tool
- GUI and CLI available
- Features: Classification, Clustering, Association
- Easy for students to explore
- Workflow: Load → Preprocess → Apply Algorithm → Analyze Results

SUMMARY

- - Data mining = finding hidden info in data
- - 8 steps called KDD Process
- - Used in shopping, banking, health, etc.
- - Helps in better decisions